



Industrial Engineering Department

INDR 262 Introduction to Optimization Methods

Term Project

- All submitted work must be entirely your own. Copying from any source is strictly prohibited.
- Any collaboration, discussion, or external source consulted in the preparation of the report must be clearly acknowledged.
- The use of AI-based tools such as ChatGPT is permitted only if explicitly declared in the submitted report. Students remain fully responsible for the accuracy, validity, and originality of all submitted content.
- Reports that are identical or substantially similar across groups will be treated as potential cases of academic dishonesty and will be subject to review.
- All students are expected to comply with Koç University's Student Code of Conduct, available at <https://apdd.ku.edu.tr/en/academic-policies/student-code-of-conduct/>

Spring 2026

Part 1: Graphical Methods and Starting Solutions (25 Pts)

Consider the following linear programming model:

$$\min Z = 16x_1 + 20x_2$$

subject to

$$-x_1 + 3x_2 \geq 0,$$

$$-5x_1 + x_2 \leq 0,$$

$$x_1 + x_2 \geq 25,$$

$$x_1, x_2 \geq 0.$$

- 1a (5 pts) Using Python and any package of your choice, plot the feasible region of the problem in two dimensions. The figure must clearly display all constraint boundaries and indicate the feasible region.
- 1b (5 pts) Determine all basic solution points of the problem. For each point, state whether it is feasible or infeasible, and report its objective-function value in a table. Using this table together with the graph, identify the optimal solution.
- 1c (5 pts) Solve only **Phase I** of the Two-Phase method by the simplex method, and provide the final simplex tableau. Do not proceed to Phase II.
- 1d (5 pts) First convert the problem into standard equality form by introducing the necessary slack, surplus, and artificial variables. Then using the standard **equality-form** constraints, formulate the **Phase I** model of the Two-Phase method in a solver. Present the mathematical formulation clearly in the report and deliver the final result obtained from this model.
- 1e (5 pts) Using the same standard equality-form constraints, formulate the Big- M model in a solver with $M = 10^4$. Present the mathematical formulation clearly in the report and report the final result obtained from this model. Note that the equality constraints are the same as in the Phase I formulation, whereas the objective function is different.

Part 2: (50 Pts)

Exam Schedule Planning

Koç University aims to prepare an examination schedule for a single day by determining classroom assignments for courses and proctor assignments for the corresponding exams. The planning task must ensure that each course is assigned to a suitable room and receives the required level of supervision. The problem integrates two linked decisions. First, each course must be assigned to one room with sufficient capacity for its enrolled students. Second, proctors must be assigned so that every exam is supervised in accordance with institutional requirements. These assignments must remain feasible with respect to proctor availability and the rule that no proctor may supervise more than one exam in the same time slot. The setting also incorporates an economic dimension. Proctor assignments generate costs, and these costs are assumed to increase with room size. In addition, larger student groups require more extensive supervision and therefore a greater number of proctors. **The following notation has to be used in the mathematical model.**

Table 1: Sets and parameters

Notation	Description
$\mathcal{C}$	set of courses to be examined
$\mathcal{R}$	set of available exam rooms
$\mathcal{P}$	set of available proctors
$\mathcal{T}$	set of exam time slots
t_c	time slot in which course $c \in \mathcal{C}$ is scheduled
K_r	seating capacity of room $r \in \mathcal{R}$
s_c	number of students enrolled in course $c \in \mathcal{C}$
a_{pt}	availability indicator, 1 available-0 not available
c_{pr}	cost of assigning proctor $p \in \mathcal{P}$ to room $r \in \mathcal{R}$
q_c	number of proctors required for course $c \in \mathcal{C}$

Table 2: Decision variables

Variable	Description
x_{cr}	equals 1 if course c is assigned to room r , and 0 otherwise
y_{pcr}	equals 1 if proctor p is assigned to course c in room r , and 0 otherwise

Koç University seeks to prepare a one-day examination plan by assigning courses to rooms and proctors to the corresponding exams. Each course must be assigned to exactly one room with sufficient capacity, and each exam must receive the required number of proctors. The plan must satisfy room-capacity limits, proctor-availability restrictions, and time-slot consistency, while minimizing total proctoring cost.

- 2a (20 pts) Formulate the problem as a mathematical optimization model. Present the complete objective function and all necessary constraints. For each, provide a one-sentence explanation stating its purpose or interpretation.
- 2b (10 pts) Solve the model using an optimization solver. Report the optimal objective value. Note that all decision variables are binary.
- 2c (5 pts) Solve the same model after relaxing the decision variables to be continuous. Do not forget to impose an upper bound of 1 on each decision variable. Does this linear programming relaxation yield the same solution as the binary model? Briefly explain why this may have occurred.
- 2d (5 pts) You are required to send emails to the teaching assistants with their exam-day assignments and to prepare the classroom door schedules showing the exams held in each room on that day. In your report, provide the exam-day schedule of TA073 and state which exam name(s) and time slot(s) should be posted on the door of SCI107.
- 2e (10 pts) There is an ongoing disciplinary evaluation involving TA018, TA055, TA040, and TA065. No pair among these four teaching assistants may be assigned to the same room for the same exam. Add the necessary constraint(s) to the model and solve it again. State in your report the additional constraints introduced by this requirement and quantify the resulting increase in the planning cost.
- 2f (5 pts) **Bonus:** Return to part (b). Suppose the dean informs you that some assignment decisions have already been made (Bonus Sheet in xlsx file), so only the remaining decisions are to be optimized. Solve the resulting model using the solver and report the new objective value together with its increase relative to the minimum-cost solution obtained in part (b).

Part 3: Knapsack Problem (25 Pts)

- 3a (5 pts) Formulate the knapsack (binary) problem clearly in your report.
- 3b (5 pts) Solve the knapsack problem under binary (0–1), integer (0, 1, 2, 3, ...), and continuous (≥ 0) decision-variable assumptions. Report all three results. Compare the resulting objective-function values and briefly comment on the differences, including which version would theoretically be expected to yield the highest objective value.
- 3c (5 pts) Derive and formulate the dual of the **LP relaxation (continuous)** of the knapsack problem, and solve this dual model.
- 3d (5 pts) Using Python, draw the feasible region of the dual model.
- 3e (5 pts) Report the shadow prices of the primal LP constraints together with the decision-variable values of the dual model, and comment on their interpretation. State the maximum amount you would be willing to pay to increase the volume capacity by one unit and the maximum amount you would be willing to pay to increase the weight capacity by one unit, and briefly comment on these results.

Submission Guidelines

- Failure to comply with the required format will result in a deduction of 10 to 30 points, depending on the extent of the deviation.
- If your code does not reproduce the reported results for a given question, you will receive zero for the coding-related part of that question.
- All groups must submit *exactly two primary files*: (i) a **PDF-formatted report** and (ii) a **single Jupyter Notebook or Python script**. Any external data files used (e.g., CSV inputs) must also be included, either separately or in a single `.zip` archive; however, modified CSV files should be submitted only if their structure was changed, which is not recommended.
- The report is expected not to exceed 15 pages, excluding the cover page.
- The PDF must contain every model, every result, and every figure required (asked) in the project.
 - Concise explanations for all parts, with figures, tables, and screenshots of Python output.
 - All sensitivity, primal, and dual results are presented clearly and consistently.
 - If a result does not appear in the PDF, it will not be graded—regardless of whether it exists in the code.
- Your code file is used only to verify that your implementation runs and reproduces the reported outputs.
 - Single Jupyter Notebook or Python script containing all implementations.
 - Include all external data files used (If assigned files are structurally modified).